

Ethics & Codes of Conduct



Most important goal

- Provides an aid to individual decision making
- Presentation addresses nine different cases (with some overlap)
 - intellectual property
 - privacy
 - confidentiality
 - professional quality
 - fairness or discrimination
 - liability
 - software risks
 - conflicts of interest
 - unauthorized access to computer systems



ACM Code of Ethics (1)

- General moral imperatives: “As an ACM member I will...”
 1. Contribute to society and human well-being.
 2. Avoid harm to others.
 3. Be honest and trustworthy.
 4. Be fair and take action not to discriminate.
 5. **Honour property rights** including copyrights and patents.
 6. Give proper credit for intellectual property.
 7. Respect the privacy of others.
 8. Honour confidentiality.



ACM Code of Ethics (2)

- Specific professional responsibilities: “As an ACM computing professional I will”:
 1. Strive to achieve the highest quality, effectiveness and dignity in both the process and products of professional work.
 2. Acquire and maintain professional competence.
 3. Know and respect existing laws pertaining to professional work.
 4. Accept and provide appropriate professional review.
 5. Give comprehensive and thorough evaluations of computer system and their impacts, including analysis of possible risks.
 6. Honour contracts, agreements, and assigned responsibilities.
 7. Improve public understanding of computing and its consequences.
 8. Access computing and communication resources only when authorized to do so.



ACM Code of Ethics (3)

- Organization leadership imperatives: “As an ACM member and an organizational leader, I will:”
 1. Articulate social responsibilities of members of an organizational unit and encourage full acceptance of those responsibilities.
 2. Manage personnel and resources to design and build information systems that enhance the quality of working life.
 3. Acknowledge and support proper and authorized uses of an organization’s computing and communication resources.
 4. Ensure that users and those who will be affected by a design have their needs clearly articulated during the assessment and design of requirements; later the system must be validated to meet requirements.
 5. Articulate and support policies that protect the dignity of users and others affected by a computing system.
 6. Create opportunities for members of the organization to learn the principles and limitations of computer systems.



ACM Code of Ethics (4)

- Compliance with the Code: “As an ACM member, I will:”
 1. Uphold and promote the principles of this Code.
 2. Treat violations of this code as inconsistent with membership in the ACM.



Ethical decision making: Case 1

- Joe is a database programmer
 - large statistical program needed by his company (actuarial requirements)
 - company programmers are encouraged to publicize their work
- Joe has found himself stuck on a problem
 - Has persisted at this for several months.
 - His manager does not recognize complexity of problem.
 - She insists job be completed in the few days.
- Joe remembers:
 - co-worker had given him source listings of their current work
 - he also has an early version of commercial software developed at another company



Ethical decision making: Case 1

- Joe studies these programs
 - Sees two areas of code which could be directly incorporated into his own program
 - He uses segments of code both from his coworker and from the commercial software
 - He does not tell anyone or mention it in the documentation.
- He completes the project and turns it in a day ahead of time.
- How does the Code of Ethics help us understand this case?



Applying the code: Case 1

- This case highlights issues involving intellectual property
 - 1.6: “Give proper credit for intellectual property”
 - Specifically, do not take credit for other’s ideas or work.
- Property rights principle (1.5)
 - copyrights, patents, trade secrets, license agreements
- Restrictions also ground in:
 - integrity (1.3)
 - complying with existing laws (2.3)



Applying the code: Case 1

- Joe violated professional ethics in two areas:
 - Failure to give credit for another's work.
 - Using code from a commercial package that (presumably) was copyrighted.
- If Joe only “looked” at co-worker's source code:
 - Could he then write his own program and still have an obligation to give credit?
- Yes:
 - He should have acknowledged credit in documentation.
 - (Some professional discretion possible here, especially if intellectual material is trivial.)



Applying the code: Case 1

- Use of commercial software code was also not appropriate:
 - Joe should have checked to determine whether or not company was authorized to use source code before using it.
- In general:
 - Desirable to share and exchange intellectual materials
 - But using bootlegged software is definitely a violation of code.



Ethical decision making: Case 2

- Three years ago, Susanne started her own consulting business
 - She is so successful she now has several people working for her.
 - Has many clients.
 - Includes work such as advising on network architectures, designing DBMSes, security.
- Presently designing a DBMS for the personnel office a medium-sized (100 person) company.
 - Susanne has involved client in design process
 - Informs CEO, CTO and human resources head about system progress



Ethical decision making: Case 2

- Now it is time to make decisions about the kind and degree of security to build into system.
- Susanne has described several options.
- Because of cost overruns, client has decided to opt for a less secure system.
 - Susanne believes information they will store is extremely sensitive (performance evaluations, medical records for insurance claims, salaries, etc.)
- With weak security:
 - Employees on workstations could figure out how to access this data.
 - Online intruders would also have access



Ethical decision making: Case 2

- Susanne feels strongly that system should be much more secure.
 - She has tried to explain the risk.
 - CEO, CTO and HR all agree that less security will do.
- What should Susanne do?
 - Should she refuse to build the system as they request?



Applying the Code: Case 2

- This case highlights issues involving privacy
 - Principle 1.7 deals with privacy
 - Principle 1.8 deals with confidentiality
- Code guidelines state that:
 - “computer professionals are obligated to preserve the integrity of data about individuals...”
 - “... from unauthorized access or accidental disclosure to inappropriate individuals”
- Code also specifies for organizational leaders:
 - Principle 3.5 (enhance personal dignity)
 - Principle 3.4 (assess needs of all those affected by system)



Applying the Code: Case 2

- Company officials:
 - Have an obligation to protect privacy of their employees.
 - Therefore they should not accept inadequate security.
- Susanne's first obligation:
 - Attempt to educate company officials (implied by principle 2.7)
- If that fails, she needs to consider her contractual obligations (principle 2.6) in honouring assigned responsibilities.
- We don't have Susanne's contract, but she may have to choose between her contract and her obligation to honour privacy and security.



Ethical Decision Making: Case 3

- Mary works in a large provincial agency dealing with alcoholism and drug abuse.
- Agency administers programs for individuals with alcohol and drug programs.
 - Maintains a large database of information on clients who use agency services.
 - Some data files contain names and current addresses of clients.
- Mary has been asked to look at the track records of treatment programs.
 - Reporting # of clients seen each month for past five years, length of client treatment, number of clients who return after program completion, criminal histories of clients.



Ethical Decision Making: Case 3

- Mary has been given access to all files in the agency's mainframe computer
 - This data is needed to put together the report.
- After assembling data:
 - She downloads it to the computer in her office.
- The agency is pressuring her to finish report on the deadline.
 - Mary decides she must work from home over the weekend.
 - She copies data onto several disks and takes them home.
 - After finishing report she leaves the disks at home and forgets about them.



Applying the Code: Case 3

- This case resembles case 2, but raises several additional issues.
- Issues involving confidentiality
 - Principle 1.7 deals with privacy
 - Principle 1.8 deals with confidentiality
- Principle 2.8 also applies:
 - Constraining access to authorized systems
- Principle 3.5:
 - Organizational leaders have obligations to “verify systems are designed and implemented to protect personal privacy and enhance personal dignity”
- Also Principle 3.3:
 - (Appropriate and authorized uses of organization’s resources)



Applying the Code: Case 3

- Government agency should have had policies and procedures to protect identity of its clients
 - Mary's friends and relatives might accidentally discover files and inappropriate uses information.
 - Note that the files Mary used did not need to have names or other information in the records.
- Agency should have removed identifying information from files Mary was allowed to use.
 - If this happened, it wouldn't have mattered that Mary copied files to her computer.
- Mary, unfortunately, was not attentive to ethical issues ahead of time.



Ethical decision making: Case 4

- Computer company is writing first part of an “efficient accounting system”.
 - Will be used by government.
 - Expectation is that this will save taxpayers a considerable amount of money each year.
- Software engineer in charge of design assigns different parts of system to his staff.
 - Reports, Internal Processing, User interface
- Manager is shown the system, and agrees it matches requirements.
- System is installed, but staff find the interface so difficult to use that their complaints are heard by upper-level management



Ethical Decision Making: Case 4

- Result of complaints:
 - upper-level management will not invest any more money in developing the new accounting system
 - they go back to their original, more expensive system



Applying the Code: Case 4

- This case highlights issues involving quality of professional work
- Code of Ethics advocates that:
 - professional strive to achieve the highest quality in both process and products (2.1)
- Principle 3.4: users and those affected by a system must have their needs clearly articulated
- Assumption in this case:
 - Failure to deliver a quality product is directly attributable to failure to follow a quality process.
 - Most likely the problems with interface could have been discovered in review process – peers or users (2.4)
- When harm results (in this case with taxpayers), failure to implement quality process clearly violates ethical behavior.



Ethical decision making: Case 5

- Contractor is determining requirements for an employment agency.
 - Client describes what is needed when displaying applications whose qualifications appear to match those for a particular job.
 - Client also further states that names of white applicants are to be displayed ahead of nonwhites.
 - Further states that names of male applicants are to be displayed ahead of female applicants.
- Recall: ethical code asserts an ACM member will be “fair and take action not to discriminate”



Applying the Code: Case 5

- This case highlights issues involving fairness and discrimination
- In this case, system designer is asked to build a system that, it appears
 - will be used to favour white males and
 - discriminate against non-whites and females
- From this it would appear that:
 - system designer should not do what he or she is told, plus
 - should also point out the problematic nature of what is being requested and ask client why this is being done
- Making the inquiry is consistent with 2.3, 2.5 and 4.1.



Applying the Code: Case 5

- If client answers that they plan to use information to favour white males, then:
 - Computer professional should refuse to build the system as proposed.
- To go ahead and build the system would violate:
 - 1.4 (fairness)
 - 2.3 (respecting existing laws)
- It would also be inconsistent with:
 - 1.1 (well-being)
 - 1.2 (avoiding harm)



Ethical decision making: Case 6

- A software development company has just produced a new software package.
 - It incorporates new tax laws and prepares both individual and small business tax returns
- The president of the company knows that the program has a number of bugs
 - He also believes the first firm to put this kind of software on the market is likely to capture the largest market share.
- The company widely advertises the package.
 - When the product is shipped, it includes a disclaimer of responsibility for errors resulting from the use of the program.



Ethical decision making: Case 6

- The company expects it will receive a number of complaints, queries, and suggestions for modification.
- The company plans to use these to make changes and eventually issue updated, improved and debugged versions.
- The president argues that this is general industry policy:
 - “Anyone who buys version 1.0 of a program knows this and will take proper precautions.”
- Because of bugs, a number of users filed incorrect tax returns and were penalized by Rev Canada.



Applying the Code: Case 6

- This case highlights issues involving legal liability for unreliable code
- Software company (and president in particular) violated several principles in the ACM code of ethics
- Since he was aware of bugs in the product, he did not strive to achieve the highest quality (Principle 2.1)
- By failing to inform consumers about bugs to system, principle 2.5 was violated.
- Here the risks to users is so great they have to pay penalties for mistakes which result from the program.
 - By law companies can make disclaimers only when they are in “good conscience” (Disclaimer does not meet legal test, violated principle 2.3)
- President also violates Principle 3.1



Ethical decision making: Case 7

- Small software company is working on an integrated inventory control system
 - very large national shoe manufacturer
 - system gathers sales data daily from stores across Canada
- Data is used by following departments:
 - accounting
 - shipping
 - ordering
- Inventory functions are critical to the smooth operation of the system and the corporation.



Ethical decision making: Case 7

- James is a quality assurance (QA) engineer with the software company
 - He suspects the inventory functions of the system are not sufficiently tested
 - However, they have passed all contracted tests.
- He is being pressured by his employers to sign off on the software.
- Legally he is only required to perform those tests which found their way into the contract
- However, his considerable experience in software testing leads him to be concerned over risks of incorrect system behavior



Ethical decision making: Case 7

- Despite insisting, James' company states:
 - “We will go out of business if we do not deliver the software on time.”
- James replies:
 - “If inventory subsystem fails, it will significantly harm our client and their employees.”
- If the potential failure were to threaten lives, it would be clear to James that he should refuse to sign off
- However, given the reduced degree of threatened harm, James is faced by a difficult decision.



Applying the Code: Case 7

- This case highlights issues involving software risks.
- Principle 1.2 stress responsibility of computing professional to avoid harm for others
 - Principle 1.1 requires concern for human well-being
 - Principle 1.3 mandates professional integrity
 - Principle 2.1 defines quality as an ethical responsibility
- These principles may conflict with agreements and commitments of an employee to the employer and client.



Applying the Code: Case 7

- The ethical imperatives of the code suggest that:
 - James should not deliver a product he believes to be inferior
 - nor should he mislead the client about the quality of the product (1.3)
- He should continue to test, and has been told of the financial repercussions of not delivering the system.
 - At the very least, the client should be informed of his reservations.



Ethical decision making: Case 8

- A software consultant is negotiating a contract with a local municipality
- Designing their traffic control system (TCS)
- She recommends they select the TCS system out of several available system on the market.
- The consultant fails to mention that she is a major stockholder of the company producing TCS software.



Applying the Code: Case 8

- This case highlights issues involving conflicts of interest.
- Principle 2.5: computer professionals must “strive to be aware, thorough and objective when evaluating, recommending and presenting system descriptions and alternatives.”
- Principle 1.3: implies a computer professional must be honest about “any circumstances that might lead to conflicts of interest”
 - IT professionals have special skills
 - It is their responsibility to ensure clients are fully aware of the options.
 - Also their responsibility to ensure professional recommendations are not modified for personal gain.



Ethical decision making: Case 9

- Hugo is working on a project for his computer science course.
- Instructor has allotted a fix amount of computer time for the project.
 - This time is enforced by the computer system.
- Hugo runs out of time, but has not yet finished the project.
- Instructor cannot be reached.



Ethical decision making: Case 9

- Hugo worked last year as a co-op programmer in the department.
- He is very familiar with procedures used to increase time allocations to accounts.
- Using what he learned last year, he is able to access the master account.
- Then he gives himself additional time.
- He now completes his project.



Applying the code: Case 9

- Principle 1.5 (property rights) has been violated.
- Principle 2.8: specifies that ACM members should “access communication resources only when authorized to do so”.
- By violating 2.8, Hugo is also violating Principle 2.3 (“know and respect existing laws”)
- As a student member of the ACM, Hugo must follow the code of ethics...
- ... even if he does not consider himself a computing professional.



Nine cases: summary

- Illustrate broad range of issues facing an IT professional.
 - Code does not prescribe what an individual must do
 - But it does identify some decisions as unacceptable.
- Ethical decision often involve balance amongst several factors.
 - Computer professionals therefore have to choose amongst the conflicting principles.
 - Goal is to ahead to the spirit as much as to the letter.



Nine cases: summary

- All of the cases also portrayed individuals acting in constrained situations.
 - Institutional environment usually influences ethical decisions.
 - Such environments can either facilitate or constrain ethical behavior.
 - Leadership roles set the tone.
 - Some of the problems shown in the cases resulting from a lack of ethical leadership.



Other applicable codes

- Canadian Information Processing Society (CIPS):
 - Code of Ethics & Standards of Conduct
- IEEE Computer Society & ACM:
 - Software Engineering Code of Ethics and Professional Practice
- Canadian Council of Professional Engineers
 - Guideline on the Code of Ethics

